

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

M. Tech. Electrical Semester – I, Examination, Dec 2013

EE703.01 Automation & Control System

Date: 03/01/2014, Friday

Time: 10.00 AM to 01.00 PM

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each section.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION I

Q - 1 Explain the radar tracking system with its block diagram in detail. Differentiate [07]
between digital controller and analog controllers.

Q - 2 (a) $G(s) = \frac{s + \frac{1}{T_1}}{s + \frac{1}{T_2}}$ [07]

Determine whether the given transfer function is a lead network or lag network for sinusoidal input $x(t) = X \sin \omega t$ and output $y_{ss}(t)$

(b) State the generalized rules for jury's stability criterion test. Determine the stability [07]
of the system through jury's stability criterion test for a given equation.

$$Q(z) = 5z^4 + 2z^3 + 3z^2 + z^1 + 19$$

OR

Q - 2 (a) Give the transient response specifications of a control system and define each. [07]
Derive the following.

1. Static position error constant
2. Static velocity error constant
3. Static acceleration error constant

(b) Apply Routh criterion to determine the stability of the system having the [07]
characteristic equation as

$$S^3 + 4 \times 10^2 S^2 + 5 \times 10^4 S + 2 \times 10^6 = 0$$

Write the z-transform of $\exp(-t)$ sampled at a frequency of 10Hz.

- Q - 3 Attempt any two [14]
- (a) Explain proportional and integral control of system in terms of steady state error [07]
- (b) $G(s) = \frac{k}{1+Ts}$ Determine steady state output $y_{ss}(t)$ for sinusoidal input $x(t) = X \sin \omega t$. Obtain z - transform of $\frac{1}{s(s+1)}$. [07]
- (c) Determine the Z-transform for [07]
- (a) Unit step (b) Ramp function (c) Exponential function (d) Sine and cosine functions

SECTION II

- Q - 4 Explain and derive the frequency response of zero order hold with its amplitude and phase plots. Define the following terms. [07]
- (a) State (b) State variables (3) State vector (d) State space
- Q - 5 (a) Sketch the polar plot for the following the following transfer function. [07]
- $G(s) = \frac{s^3}{(s+1)(s+2)}$ Give advantages and limitations of frequency response analysis.
- (b) Sketch Bode plot for a given transfer function. $\frac{50K}{s(s+10)(s+5)(s+1)}$ [07]

OR

- Q - 5 (a) Consider a unity feedback control system having the open loop transfer function. [07]
- $G(s)H(s) = \frac{k(s+2)(s+1)}{(s+0.1)(s-1)}$
- Draw the root locus of the system with the gain K as a variable. Find from the root locus plot, the value of gain K for which a closed-loop system is critically damped.
- (b) Explain the effect of addition of open-loop poles and zeros in root locus plot. [07]
- Explain the advantages of root locus method

- Q - 6 Attempt any two [14]
- (a) Explain under damped, over damped and critically damped cases for second order system in detail [07]

- (b) Check for controllability and observability of a system having following coefficient [07]
matrices

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, C^T = \begin{bmatrix} 10 \\ 5 \\ 1 \end{bmatrix}$$

- (c) Determine the transfer function from the data given below:- [07]

$$(a) \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} u$$

$$\text{and } y = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(b) \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -25 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 25 \\ -20 \end{bmatrix} u$$

$$\text{and } y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$
